

LLDD BE - Job Batch Email and SRM Integration

SBP Mall - ระบบประกันรายได้ | Low Level Design Document

แนวทางการใช้เอกสาร

หัวข้อ	คำอธิบาย
วัตถุประสงค์	ใช้เป็นรายละเอียดระดับพัฒนาสำหรับออกแบบ ลงมือพัฒนา ตรวจสอบ และทดสอบขอบเขตที่ระบุในฉบับนี้
ลำดับการอ่าน	เริ่มจาก Overview และ Scope จากนั้นตรวจ Field/Validation, Implementation, Contract, Processing Flow, Acceptance Criteria และ Test Checklist ตามลำดับ
ผลลัพธ์ที่คาดหวัง	ผู้อ่านสามารถระบุ input, ขั้นตอนประมวลผล, output, เงื่อนไขผิดพลาด และหลักฐานการทดสอบได้จากเนื้อหาในฉบับเดียว

คำว่า Input, Progress และ Output ในเอกสารนี้หมายถึงข้อมูลตั้งต้น ลำดับการทำงาน และผลลัพธ์ที่ตรวจสอบได้ของขอบเขตที่กำลังพัฒนา

1. Overview

รายการ	รายละเอียด
Track	BE
Estimate	24 ชั่วโมง
Owner	Butsaba <But> Podamrong
Objective	ออกแบบ Backend contracts สำหรับ Batch Job Admin, interface tracking/pending ACK, Email Template Service และ SRM Integration Adapter

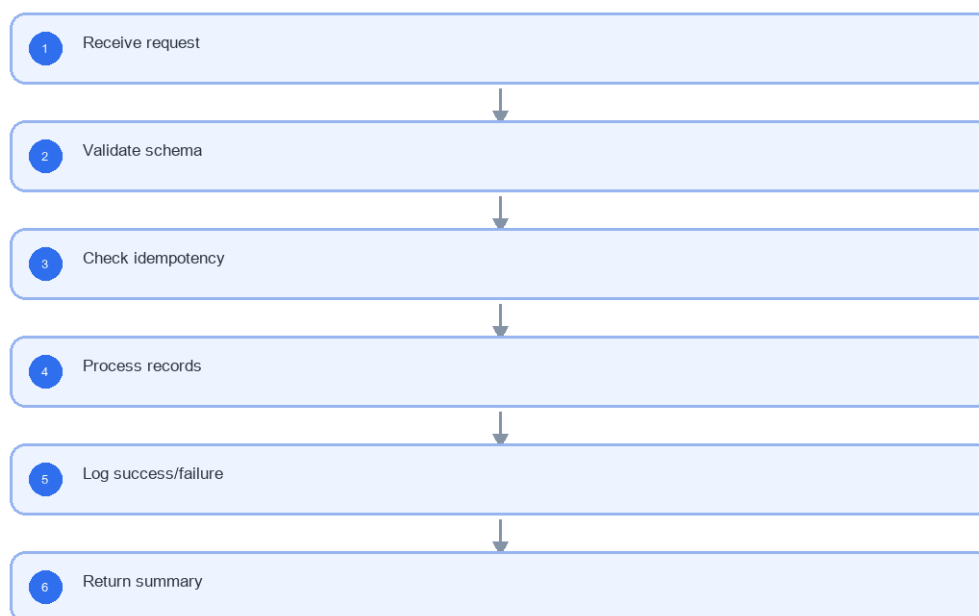
Common contract reference: ทุกหัวข้อ API/FE ต้องยึด LLDD-BE-API-Common-Contracts.pdf และ LLDD-FE-Integration-Contracts.pdf สำหรับ error/auth/format/pagination/action/RBAC ก่อนลงรายละเอียดเฉพาะหน้าหรือเฉพาะ endpoint

2. Screen / Functional Scope

- Batch Job Admin APIs ครบ 6 endpoints
- Interface tracking และ pending ACK APIs
- Job runner guard/history
- Notification adapter
- STA ACK callback
- SRM integration inbound is optional/reference

4. Implementation Flow Diagram (Reference)

LLDD BE - Job Batch Email and SRM Integration



รูปที่ 1: Implementation flow reference: LLDD BE - Job Batch Email and SRM Integration

5. Field, Format, and Validation

Field / UI	Format	Validation	Behavior
jobNo	string	required	maps to job registry
sourceRefNo	string	required for SRM	idempotency key
templateCode	EM-xx	required	email template key
transactionId	uuid	generated	integration log key

5.1 Input / Progress / Output Contract

Stage	Contract for implementation
Input	GET /api/v1/jobs; GET /api/v1/jobs/{jobNo}; PUT /api/v1/jobs/{jobNo}/params
Progress	Receive request; Validate schema; Check idempotency; Process records
Output	job_configs; job_run_histories; interface_transactions

5.90 Endpoint Implementation Contract

Endpoint	Use-case owner	Service/repository behavior	Definition of done
GET /api/v1/jobs	รายการ 11 job entry points พร้อมสถานะล่าสุด	Receive request	job run guard prevents duplicate running job
GET /api/v1/jobs/{jobNo}	รายละเอียด job และ typed parameter metadata	Validate schema	SRM duplicate skipped
PUT /api/v1/jobs/{jobNo}/params	บันทึกเฉพาะ parameter key ที่ metadata ระบุ editable	Check idempotency	email preview renders variables
PUT /api/v1/jobs/{jobNo}/enabled	เปิด/ปิด job พร้อม audit reason	Process records	failed records include detail
POST /api/v1/jobs/{jobNo}/run	สั่ง manual run/retry โดยตรวจสอบ enabled และ concurrent run	Log success/failure	job run guard prevents duplicate running job
GET /api/v1/jobs/{jobNo}/runs	ประวัติการรันแบบแบ่งหน้า	Return summary	SRM duplicate skipped
GET /api/v1/interfaces/tracking	ค้นสถานะ interface ตาม dataset/business key/status/ช่วงเวลา	Receive request	email preview renders variables
GET /api/v1/interfaces/pending-ack	รายการ ACK ค้างตาม watchdog rule อายุอย่างน้อย 1 วัน	Validate schema	failed records include detail
GET /api/v1/email-templates/{code}	อ่าน template และ merge-variable metadata	Check idempotency	job run guard prevents duplicate running job
PUT /api/v1/email-templates/{code}	บันทึก subject/body	Process records	SRM duplicate skipped
POST /api/v1/email-templates/{code}/preview	render template โดยไม่บันทึก	Log success/failure	email preview renders variables
POST /api/v1/email-templates/{code}/reset	รีเซ็ต template รายตัว	Return summary	failed records include detail
POST /api/v1/interfaces/sta/ack	STA ACK callback ให้ Job 10 เป็น safety net	Receive request	job run guard prevents duplicate running job
POST /api/v1/integrations/srm/income-guarantee	รับข้อมูลประกันรายได้จาก SRM	Validate schema	SRM duplicate skipped

5.91 Backend Execution Sequence

Step	Behavior specific to this LLDD	Failure/test evidence
1	Receive request	run job
2	Validate schema	run duplicate
3	Check idempotency	SRM missing field
4	Process records	SRM duplicate
5	Log success/failure	email preview
6	Return summary	template save

6. Button / User Action Mapping

Action	Trigger	API / Service	Expected Result
Run job	POST	jobRunner.run	queued/run history
Receive SRM	POST	srmlIntegration.ingest	transaction result
Preview email	POST	emailTemplate.render	merged subject/body

7. API Contract

GET /api/v1/jobs

รายการ 11 job entry points พร้อมสถานะล่าสุด

Query Params
<pre>{ "page": 1, "size": 20 }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response
<pre>{ "page": 1, "size": 20, "total": 11, "items": [{ "jobNo": "8b", "name": "StartInternalWorkflow", "enabled": true, "scheduleMode": "AFTER_JOB", "scheduleExpression": "8", "currentStatus": "SUCCESS", "lastRunId": 4451, "lastRunAt": "2026-07-22T05:35:00+07:00" }] }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].name	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].enabled	boolean	Yes	UTF-8; use value domain described by endpoint purpose
items[].scheduleMode	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].scheduleExpression	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].currentStatus	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].lastRunId	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].lastRunAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null

GET /api/v1/jobs/{jobNo}

รายละเอียด job และ typed parameter metadata

Query Params
<pre>{ "jobNo": "8b" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
jobNo	string	No	UTF-8; use value domain described by endpoint purpose

Response
<pre>{ "jobNo": "8b", "name": "StartInternalWorkflow", "enabled": true, "scheduleMode": "AFTER_JOB", "scheduleExpression": "8", "parameters": [{ "key": "period", "label": "งวดข้อมูล", "type": "MONTH", "value": "2026-07", "editable": true, "required": true }] }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
name	string	Yes	UTF-8; use value domain described by endpoint purpose
enabled	boolean	Yes	UTF-8; use value domain described by endpoint purpose
scheduleMode	string	Yes	UTF-8; use value domain described by endpoint purpose
scheduleExpression	string	Yes	UTF-8; use value domain described by endpoint purpose
parameters	array<object>	Yes	JSON array; element type shown in Type column
parameters[].key	string	Yes	UTF-8; use value domain described by endpoint purpose
parameters[].label	string	Yes	UTF-8; use value domain described by endpoint purpose
parameters[].type	string	Yes	UTF-8; use value domain described by endpoint purpose
parameters[].value	string	Yes	UTF-8; use value domain described by endpoint purpose
parameters[].editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose
parameters[].required	boolean	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/jobs/{jobNo}/params

บันทึกเฉพาะ parameter key ที่ metadata ระบุ editable

Request
<pre>{ "params": { "period": "2026-07" }, "reason": "ปรับงวด rerun" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
params	object	Yes	JSON object; nested fields listed below
params.period	string	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "jobNo": "8b",
  "configVersion": 12,
  "updatedKeys": [
    "period"
  ],
  "message": "saved"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
configVersion	integer	Yes	UTF-8; use value domain described by endpoint purpose
updatedKeys	array<string>	Yes	JSON array; element type shown in Type column
message	string	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/jobs/{jobNo}/enabled

เปิด/ปิด job พร้อม audit reason

Request

```
{
  "enabled": false,
  "reason": "ปิดชั่วคราวช่วงปิดงบ"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
enabled	boolean	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "jobNo": "8b",
  "enabled": false,
  "message": "saved"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
enabled	boolean	Yes	UTF-8; use value domain described by endpoint purpose
message	string	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/jobs/{jobNo}/run

สั่ง manual run/retry โดยตรวจ enabled และ concurrent run

Request

```
{
  "params": {
    "period": "2026-07"
  },
  "reason": "rerun หลังแก้ไขข้อมูล"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
params	object	Yes	JSON object; nested fields listed below
params.period	string	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "runId": 4452,
  "jobNo": "8b",
  "status": "QUEUED",
  "queuedAt": "2026-07-22T11:00:00+07:00"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
runId	integer	Yes	UTF-8; use value domain described by endpoint purpose
jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
status	string	Yes	UTF-8; use value domain described by endpoint purpose
queuedAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null

GET /api/v1/jobs/{jobNo}/runs

ประวัติการรันแบบแบ่งหน้า

Query Params

```
{
  "status": "FAILED",
  "page": 1,
  "size": 20
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
status	string	No	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "items": [
    {
      "runId": 4450,
      "jobNo": "8b",
      "status": "FAILED",
      "triggerType": "MANUAL",
      "triggeredBy": "E001",
      "startedAt": "2026-07-22T05:20:00+07:00",
      "endedAt": "2026-07-22T05:21:30+07:00",
      "durationSec": 90,
      "readCount": 10,
      "successCount": 9,
      "rejectCount": 1,
      "errorCode": "GEN_FLOW_GATE_NOT_READY",
      "errorMessage": "ข้อมูลผู้อนุมัติยังไม่ครบ"
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].runId	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].jobNo	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].status	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].triggerType	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].triggeredBy	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].startedAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null
items[].endedAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null
items[].durationSec	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].readCount	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].successCount	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].rejectCount	integer	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
items[].errorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].errorMessage	string	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/interfaces/tracking

ค้นสถานะ interface ตาม dataset/business key/status/ช่วงเวลา

Query Params

```
{
  "dataName": "COMPENSATE_INIT_I",
  "status": "SENT",
  "pending": true,
  "sentFrom": "2026-07-01T00:00:00+07:00",
  "sentTo": "2026-07-22T23:59:59+07:00",
  "page": 1,
  "size": 20
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
dataName	string	No	UTF-8; use value domain described by endpoint purpose
status	string	No	UTF-8; use value domain described by endpoint purpose
pending	boolean	No	UTF-8; use value domain described by endpoint purpose
sentFrom	string	No	UTF-8; use value domain described by endpoint purpose
sentTo	string	No	UTF-8; use value domain described by endpoint purpose
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "items": [
    {
      "trackingId": 9912,
      "dataName": "COMPENSATE_INIT_I",
      "direction": "OUT",
      "businessKey": "2569/00098",
      "docNo": "2569/00098",
      "fileName": "COMPENSATE_INIT_I_25690722.dat",
      "status": "SENT",
      "sentAt": "2026-07-20T17:02:00+07:00",
      "ackedAt": null,
      "returnCode": null,
      "ageHours": 41
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].trackingId	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].dataName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].direction	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].businessKey	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].docNo	string	Yes	พ.ศ. YYYY/xxxxx
items[].fileName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].status	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].sentAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null
items[].ackedAt	string null	No	ISO-8601 ค.ศ.; nullable only when type includes null
items[].returnCode	string null	No	UTF-8; use value domain described by endpoint purpose
items[].ageHours	integer	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/interfaces/pending-ack

รายการ ACK ค้างตาม watchdog rule อายุอย่างน้อย 1 วัน

Query Params
<pre>{ "thresholdHours": 24, "dataName": "COMPENSATE_INIT_I", "page": 1, "size": 20 }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
thresholdHours	integer	No	UTF-8; use value domain described by endpoint purpose
dataName	string	No	UTF-8; use value domain described by endpoint purpose
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "count": 1,
  "items": [
    {
      "trackingId": 9912,
      "dataName": "COMPENSATE_INIT_I",
      "businessKey": "2569/00098",
      "docNo": "2569/00098",
      "fileName": "COMPENSATE_INIT_I_25690722.dat",
      "sentAt": "2026-07-20T17:02:00+07:00",
      "ageHours": 41,
      "returnCode": null
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
count	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].trackingId	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].dataName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].businessKey	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].docNo	string	Yes	พ.ศ. YYYY/xxxxx
items[].fileName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].sentAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null
items[].ageHours	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].returnCode	string null	No	UTF-8; use value domain described by endpoint purpose

GET /api/v1/email-templates/{code}

อ่าน template และ merge-variable metadata

Query Params

```
{
  "code": "EM-01"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
code	string	No	UTF-8; use value domain described by endpoint purpose

Response

```
{
  "code": "EM-01",
  "name": "แจ้งสร้างเอกสาร",
  "subject": "เอกสาร {{docNo}}",
  "body": "กรุณาดูตรวจสอบเอกสาร {{docNo}}",
  "variables": [
    "docNo"
  ],
  "fromRule": "SYSTEM",
  "toRule": "NEXT_SECTION",
  "ccRule": "NONE"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
code	string	Yes	UTF-8; use value domain described by endpoint purpose
name	string	Yes	UTF-8; use value domain described by endpoint purpose
subject	string	Yes	UTF-8; use value domain described by endpoint purpose
body	string	Yes	UTF-8; use value domain described by endpoint purpose
variables	array<string>	Yes	JSON array; element type shown in Type column
fromRule	string	Yes	UTF-8; use value domain described by endpoint purpose
toRule	string	Yes	UTF-8; use value domain described by endpoint purpose
ccRule	string	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/email-templates/{code}

บันทึก subject/body

Request

```
{
  "subject": "แจ้งเอกสาร {{docNo}}",
  "body": "กรุณาตรวจสอบเอกสาร {{docNo}}",
  "reason": "ปรับลดค่า"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
subject	string	Yes	UTF-8; use value domain described by endpoint purpose
body	string	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "code": "EM-01",
  "subject": "แจ้งเอกสาร {{docNo}}",
  "body": "กรุณาตรวจสอบเอกสาร {{docNo}}",
  "updatedAt": "2026-07-22T10:20:00+07:00"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
code	string	Yes	UTF-8; use value domain described by endpoint purpose
subject	string	Yes	UTF-8; use value domain described by endpoint purpose
body	string	Yes	UTF-8; use value domain described by endpoint purpose
updatedAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null

POST /api/v1/email-templates/{code}/preview

render template โดยไม่บันทึก

Request

```
{
  "variables": {
    "docNo": "2569/00123"
  }
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
variables	object	Yes	JSON object; nested fields listed below
variables.docNo	string	Yes	พ.ศ. YYYY/xxxxx

Response

```
{
  "subject": "แจ้งเอกสาร 2569/00123",
  "body": "กรุณาตรวจสอบเอกสาร 2569/00123"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
subject	string	Yes	UTF-8; use value domain described by endpoint purpose
body	string	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/email-templates/{code}/reset

รีเซ็ต template รายตัว

Request
<pre>{ "reason": "สินค้าเริ่มต้น" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response
<pre>{ "code": "EM-01", "reset": true }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
code	string	Yes	UTF-8; use value domain described by endpoint purpose
reset	boolean	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/interfaces/sta/ack

STA ACK callback ให้ Job 10 เป็น safety net

Request
<pre>{ "transactionId": "TX-001", "returnCode": "A", "receivedAt": "2026-07-20T10:00:00+07:00" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
transactionId	string	Yes	UTF-8; use value domain described by endpoint purpose
returnCode	string	Yes	UTF-8; use value domain described by endpoint purpose
receivedAt	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null

Response
<pre>{ "message": "acknowledged" }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
message	string	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/integrations/srm/income-guarantee

รับข้อมูลประกันรายได้จาก SRM

Request
<pre>{ "sourceSystem": "SRM", "sourceRefNo": "SRM-001", "impactedStoreCode": "01234", "periodMonth": "2569-07" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
sourceSystem	string	Yes	UTF-8; use value domain described by endpoint purpose
sourceRefNo	string	Yes	UTF-8; use value domain described by endpoint purpose
impactedStoreCode	string	Yes	exactly 5 digits; preserve leading zero
periodMonth	string	Yes	ISO-8601 ค.ศ.; nullable only when type includes null

Response
<pre>{ "transactionId": "TX-001", "successRecords": 1, "failedRecords": 0 }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
transactionId	string	Yes	UTF-8; use value domain described by endpoint purpose
successRecords	integer	Yes	UTF-8; use value domain described by endpoint purpose
failedRecords	integer	Yes	UTF-8; use value domain described by endpoint purpose

8. Reference DB Mapping (No Database Page Work)

ส่วนนี้เป็นข้อมูลอ้างอิงสำหรับการ implement API/Job เท่านั้น ไม่ใช้งานสร้างหน้า Database, ไม่ใช้งานออกแบบ DB page และไม่ถูกนับเป็น deliverable แยกของ FE/BE

Table / Object	R/W	Usage
job_configs	R/W	enabled, cron, params ของ batch
job_run_histories	R/W	ประวัติการรันและสถานะล่าสุด
interface_transactions	R/W	tracking file/API interface และ ACK

Table / Object	R/W	Usage
email_templates	R/W	subject/body template
status_email_rules	R	TO/CC ตามสถานะ
audit_logs	R/W	audit การแก้ job/email

9. Processing Flow

Step	Description
1	Receive request
2	Validate schema
3	Check idempotency
4	Process records
5	Log success/failure
6	Return summary

10. Acceptance Criteria

- job run guard prevents duplicate running job
- SRM duplicate skipped
- email preview renders variables
- failed records include detail

11. Developer Test Checklist

No	Test
1	run job
2	run duplicate
3	SRM missing field
4	SRM duplicate
5	email preview
6	template save